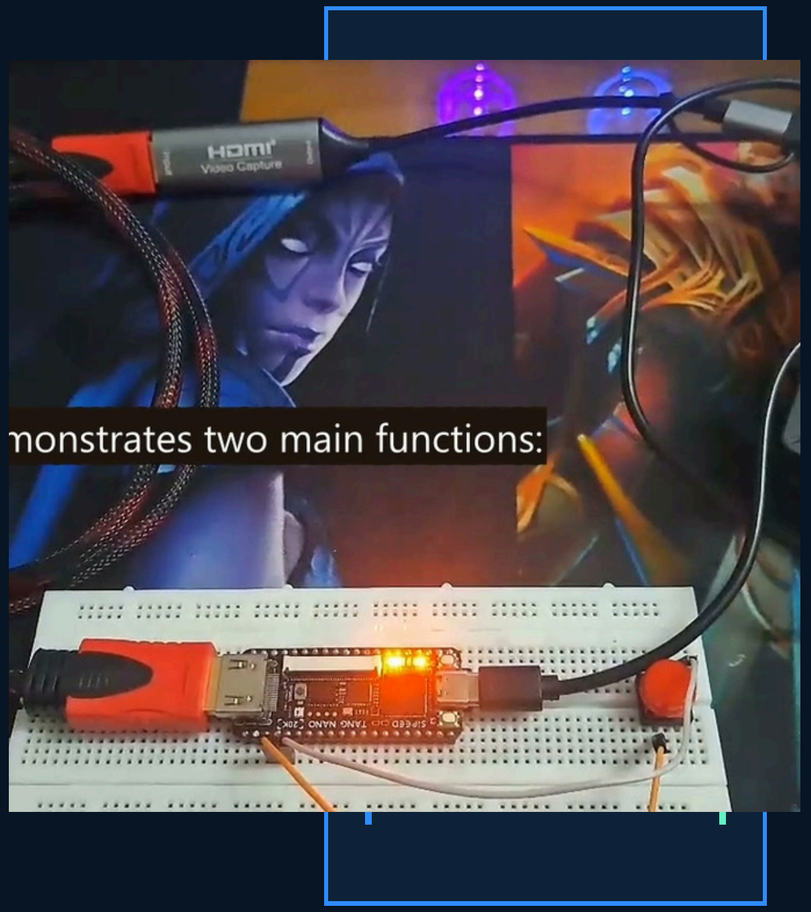
RISC-V Controlled INT8 CNN Accelerator on FPGA



Tang Nano 20K • RV32I • INT8 CNN • microSD • UART • HDMI

Group 7 | Ho Chi Minh City University of Technology and Engineering | October 2026

From a Fixed Demo to a Reloadable Edge AI System

The solution separates fixed hardware, reloadable firmware and replaceable AI data.



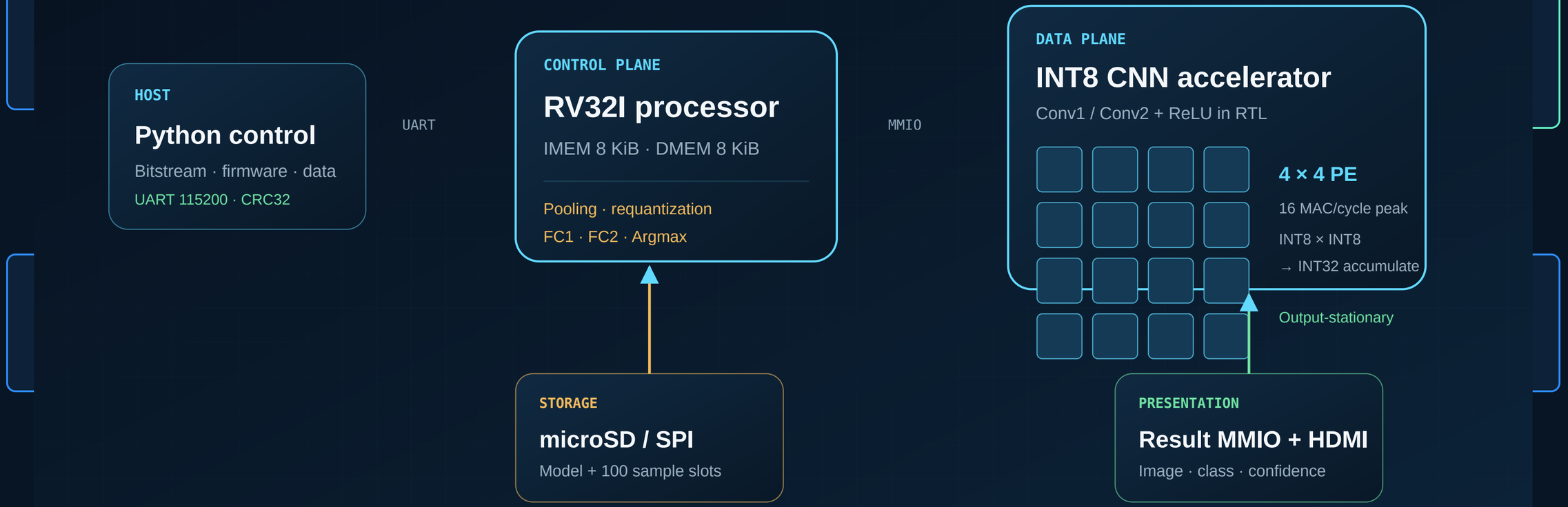
Rebuild only when the FPGA hardware changes.

Programs, weights and inputs can change without place-and-route.

System Architecture

IMPLEMENTED SYSTEM ARCHITECTURE

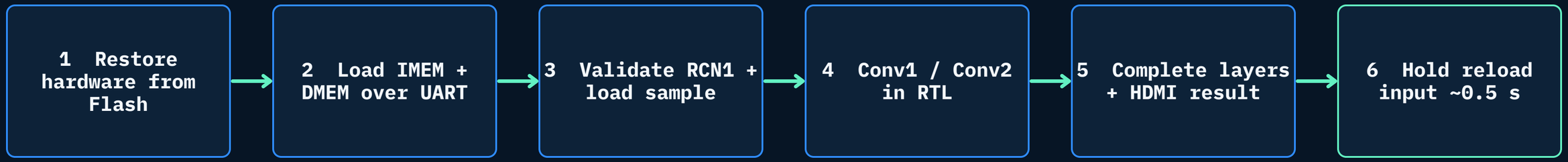
Tang Nano 20K · RV32I · signed INT8 convolution · firmware-managed I/O



Verified at RTL block, tensor, firmware/RTL integration, and FPGA implementation levels.

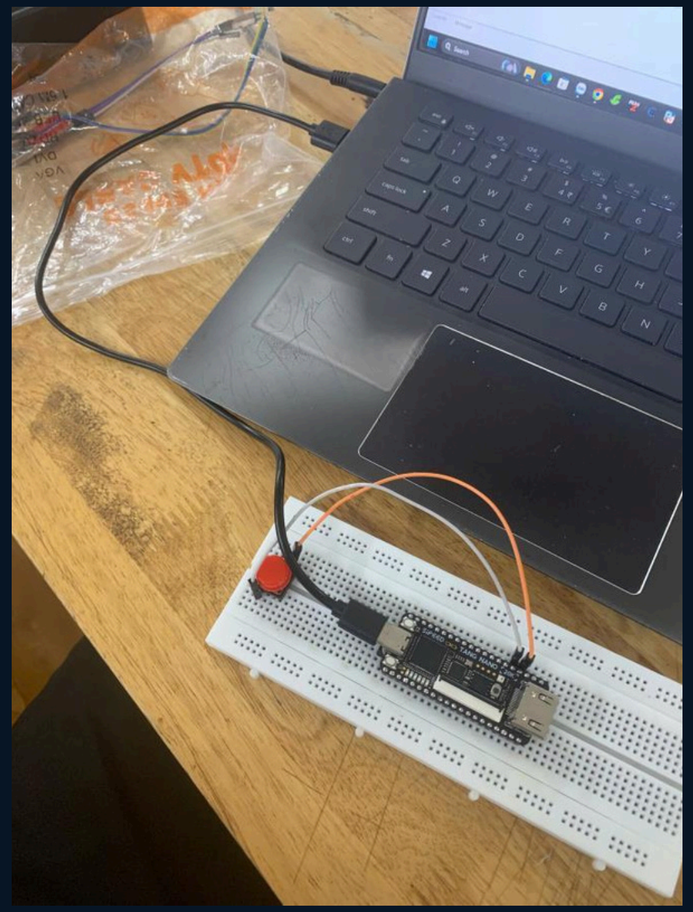
Boot, Inference and Reload Sequence

A fixed FPGA image supports a repeatable boot, inference and firmware-reload path.



Control-plane reload ≠ FPGA reconfiguration

RV32I Core and UART-Loaded Instruction Memory



```
[PASS] Test 1: Arithmetic and Logic
[PASS] Test 2: Shift and Comparison
[PASS] Test 3: Data Memory
[PASS] Test 4: Branch
[PASS] Test 5: LUI and AUIPC
[PASS] Test 6: JAL and JALR

All tests completed.
Passed: 6/6
All 6 test groups PASSED.
```

Standalone validation build
1 KiB instruction memory
133 instructions, 532 bytes
UART at 115,200 baud
6 / 6 instruction groups passed

Integrated AI SoC uses separate 8 KiB IMEM and 8 KiB DMEM.

LeNet-Inspired INT8 Inference Pipeline

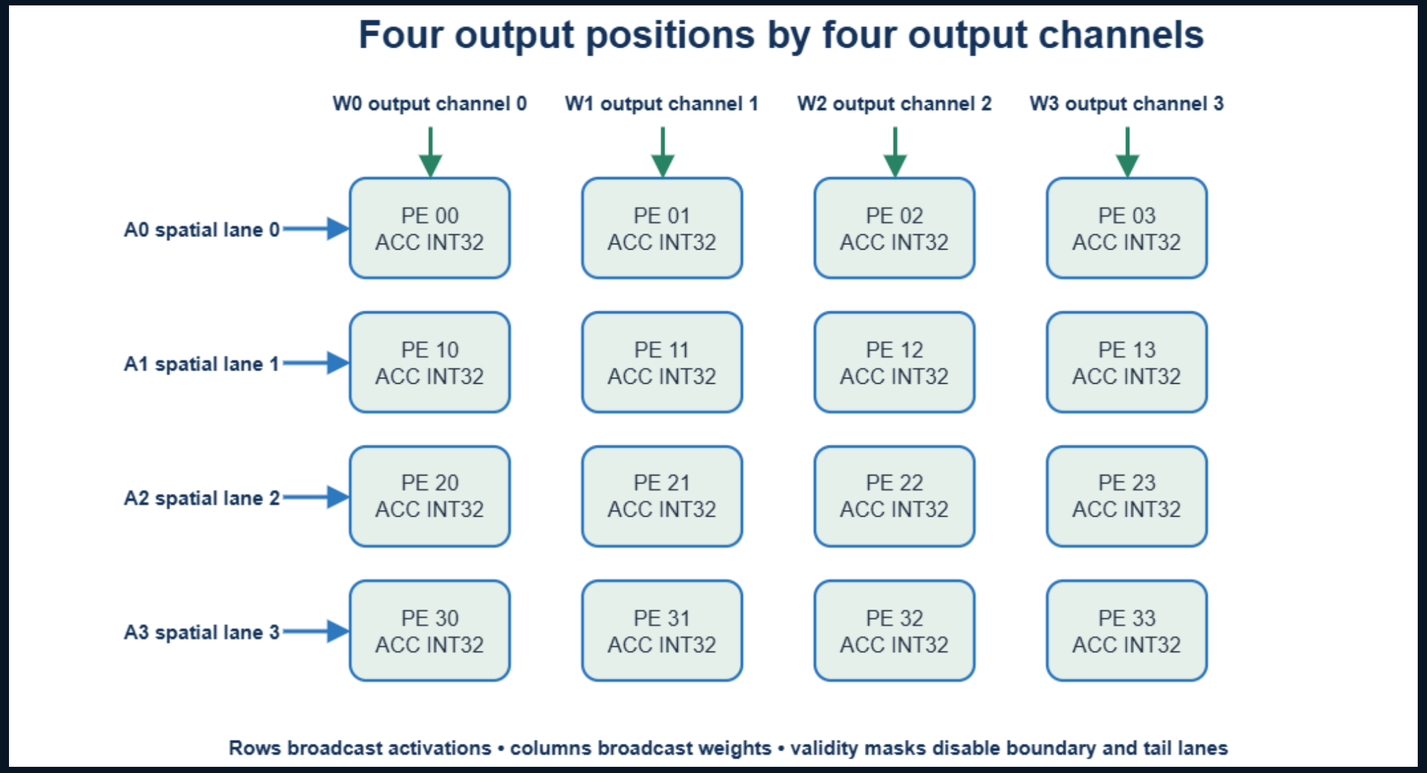
Signed INT8 activations and weights, INT16 products and INT32 accumulators.



Conv1 and Conv2 run in RTL. Pooling, requantization, FC1, FC2 and Argmax run in RV32I firmware. The accelerated convolutions represent 83.4% of learned-layer MACs, not a measured speedup.

4 × 4 Convolution Accelerator

Sixteen PEs map four output positions by four output channels with boundary masks.



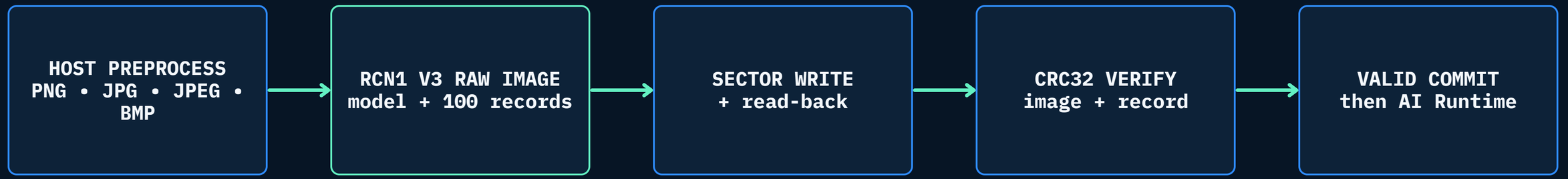
Peak: 16 signed INT8 MACs/cycle

4 activation lanes (row broadcast)
4 weight lanes (column broadcast)

Local buffers
3,136 INT8 inputs
4,608 INT8 weights
12,544 INT32 outputs

States: IDLE • CHECK • RUN • DONE • ERROR

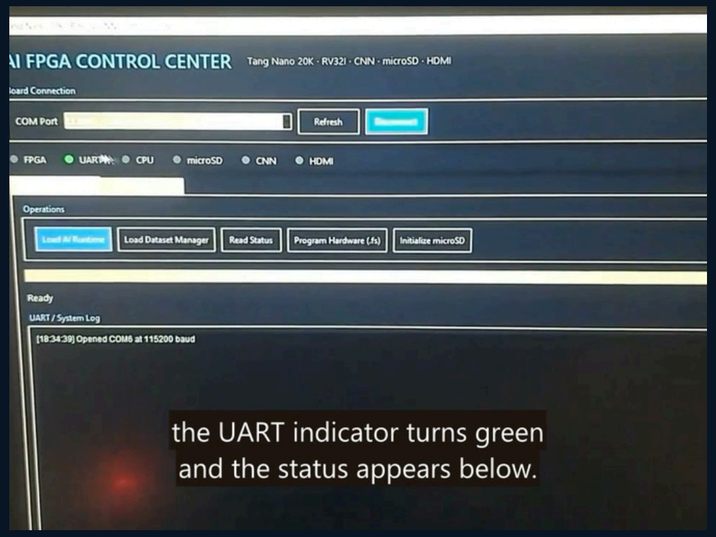
microSD Data and Firmware Workflow

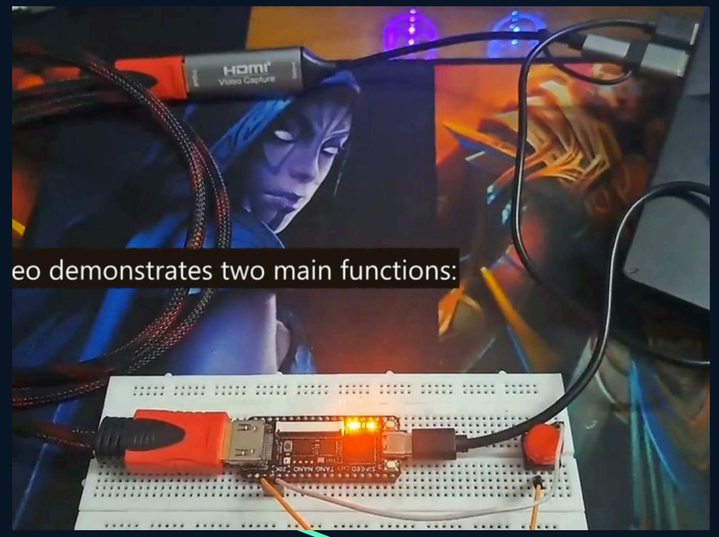


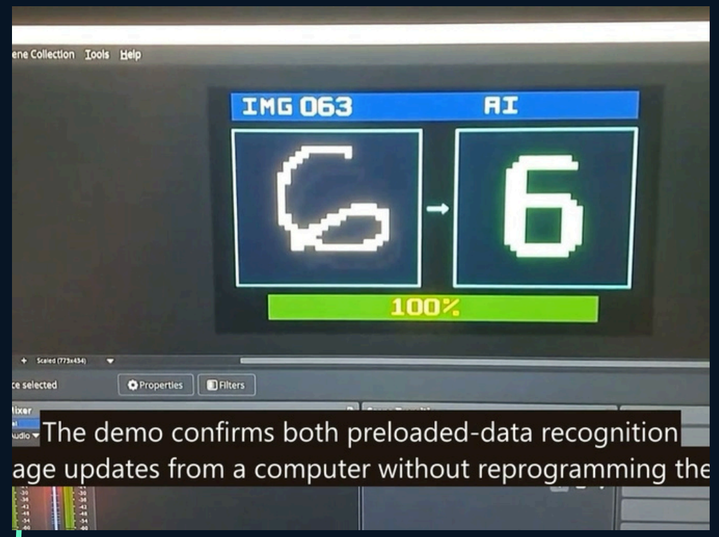
290,816 bytes • 568 sectors • 100 fixed sample slots • incomplete updates never become valid

Integrated Board Demonstration

A repeatable host-to-board workflow with real UART, microSD and HDMI evidence.







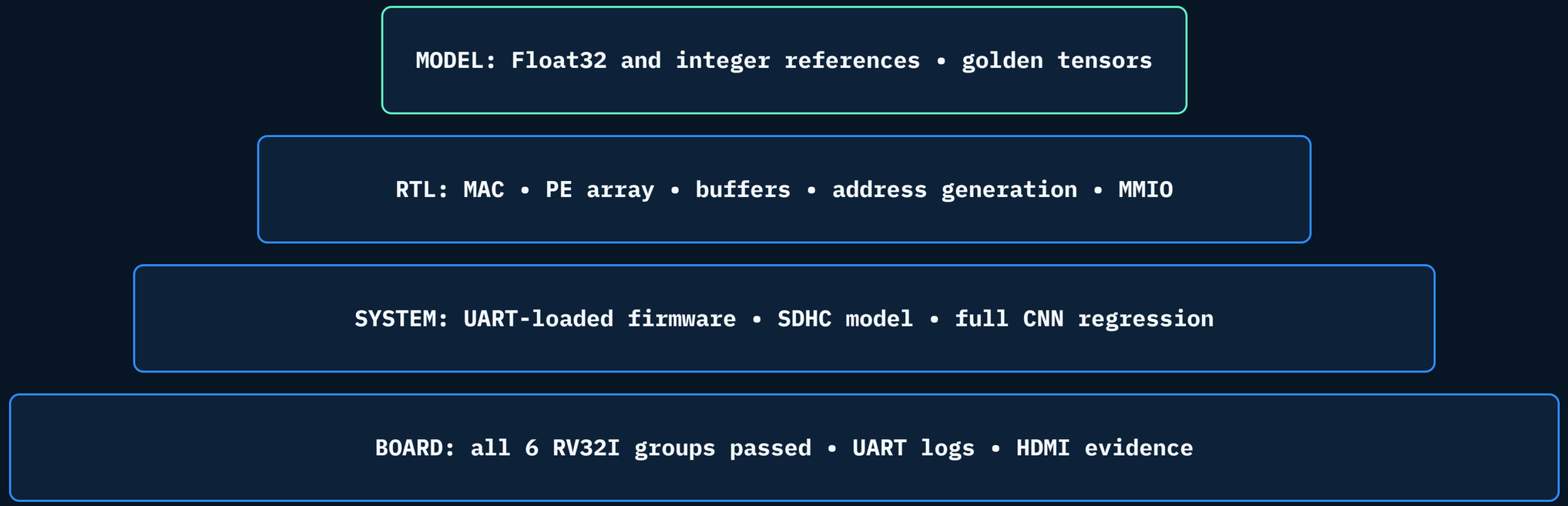
CONTROL CENTER
connect • program • load

TANG NANO 20K
USB-UART • microSD • HDMI

HDMI RESULT
image • class • confidence

Verification Strategy

Verification progresses from arithmetic and tensor references to full firmware and board behavior.

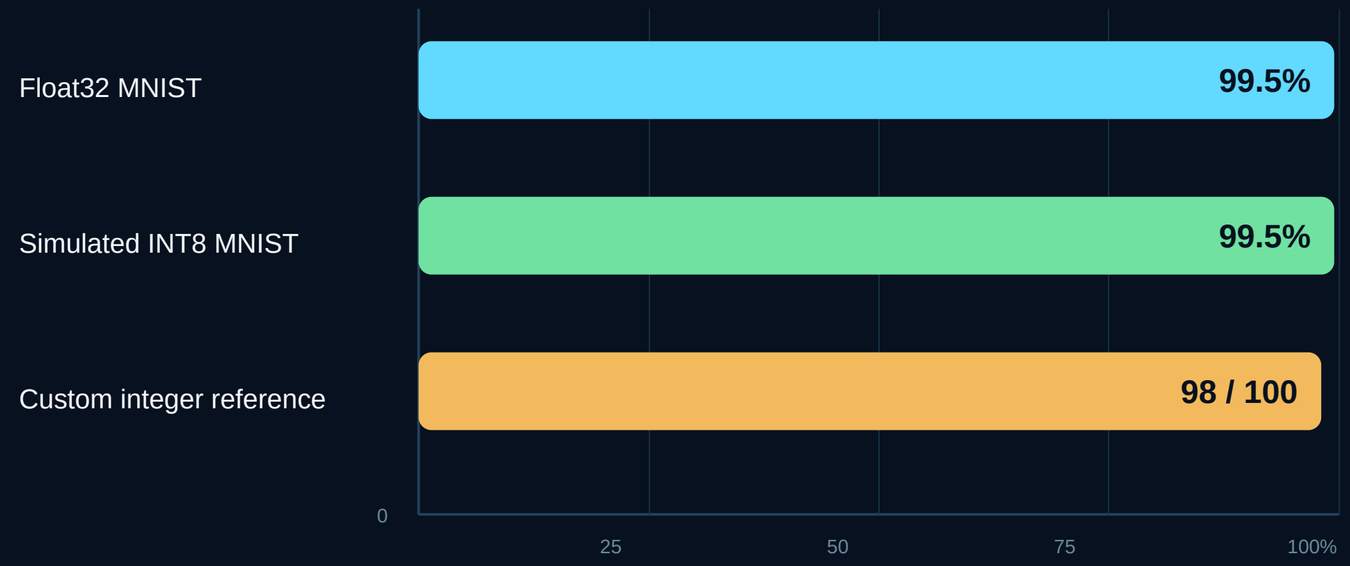


Verified Model Results

Only recorded model and integer-reference results are presented as quantitative evidence.

Verified accuracy evidence

Offline model and exported-integer reference results



Physical-board accuracy across all 100 records has not yet been measured.

NOT YET RECORDED

Physical-board accuracy across all 100 samples
End-to-end latency
Software-only RV32I baseline
Measured accelerator speedup

These results are intentionally not claimed.

FPGA Implementation and Timing

Final Gowin post-place-and-route evidence for the Tang Nano 20K target.

Gowin PnR Analysis Report

Tang Nano 20K · GW2AR-LV18QN88C8/I7 · Gowin EDA V1.9.11.03 Education · 17 October 2026

Resource Usage Summary

Resource	Usage	Utilization
Logic	15715/20736	76%
--LUT, ALU, ROM16	15475 (11356 LUT, 4119 ALU, 0 ROM16)	–
--SSRAM (RAM16)	40	–
Register	3264/15750	21%
--Logic Register as Latch	0/15552	0%
--Logic Register as FF	3259/15552	21%
--I/O Register as Latch	0/198	0%
--I/O Register as FF	5/198	3%
CLS	8870/10368	86%
I/O Port	26/66	40%
I/O Buffer	22 (6 input, 16 output, 0 bidirectional)	–
IOLOGIC	4 OSER10	7%
BSRAM	46 SDPB	100%
DSP	2 MULT18X18, 2 MULTALU36X18, 19 MULTADDALU18X18	92%

Source: final post-place-and-route report, TangNano20K_RISCV_CNN.rpt.html.

Gowin Timing Analysis Summary

Tang Nano 20K · GW2AR-LV18QN88C8/I7 · Gowin EDA V1.9.11.03 Education · 17 October 2026

Static Timing Analysis Run Summary

Setup Model	Hold Model	Paths / Endpoints	Violated Endpoints
Slow 0.95 V, 85 °C, C8/I7	Fast 1.05 V, 0 °C, C8/I7	12,675 paths / 12,435 endpoints	Setup: 0 · Hold: 15

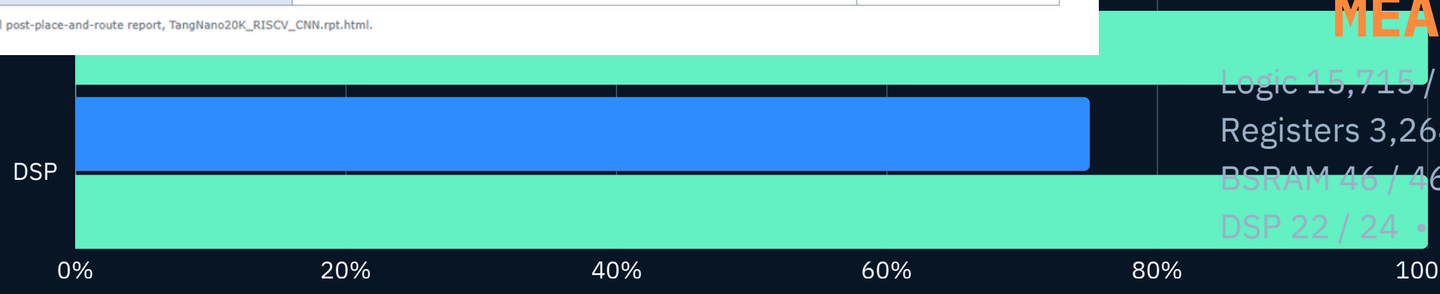
Clock Constraints and Achieved Maximum Frequency

Clock	Constraint	Actual Fmax	Logic Level	Timing Margin
sys_clk	27.000 MHz	137.825 MHz	10	Pass
soc_clk	5.400 MHz	6.617 MHz	109	Pass
pixel_clk	27.000 MHz	33.803 MHz	24	Pass

Representative Worst Setup and Hold Paths

Type	Slack (ns)	From Node	To Node	Clock Domain
Setup #1	+7.454	u_hdmi/u_renderer/v_cnt_4_s0/Q	u_hdmi/u_renderer/rgb_b_3_s0/D	pixel_clk → pixel_clk
Hold #1	−1.435	u_soc/u_bootloader/run_sample_4_s0/Q	boot_run_sample_meta_4_s0/D	soc_clk → sys_clk
Hold #2	−1.420	u_soc/u_bootloader/run_sample_6_s0/Q	boot_run_sample_meta_6_s0/D	soc_clk → sys_clk
Hold #15	−1.137	u_soc/u_bootloader/cpu_release_s0/Q	boot_release_sys_0_s0/D	soc_clk → sys_clk

Source: final post-place-and-route timing report, TangNano20K_RISCV_CNN_tr_content.html.



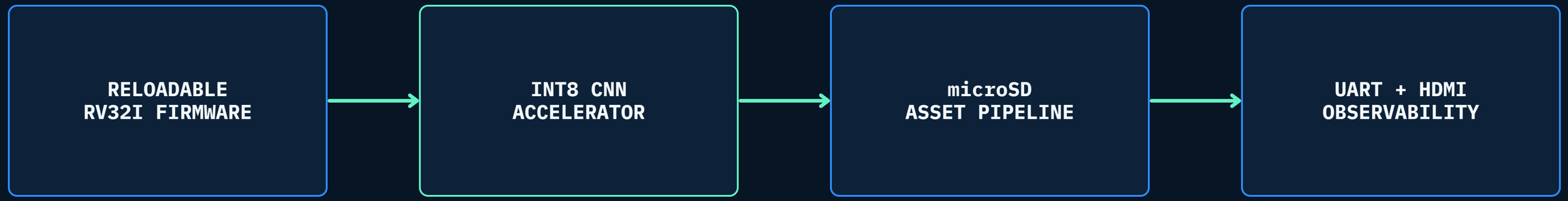
MEASURED

Logic 15,715 / 20,736 • 76%
Registers 3,264 / 15,750 • 21%
BSRAM 46 / 46 • 100%
DSP 22 / 24 • 92%

Setup passes
SoC target 5.4 MHz
Reported Fmax 6.617 MHz

15 hold-violated endpoints remain

What the Project Delivers



The project combines RTL, C firmware, Python host tools, INT8 model artifacts, verification environments and physical-board evidence. Next priorities: hold closure, cycle counters, a version-matched 100-sample board log and a software-only RV32I baseline.

Project Repository

Source code, technical documentation, implementation evidence and project demonstrations.

GITHUB REPOSITORY

https://github.com/NguyenKT0911/Embedded-system-final

YOUTUBE PLAYLIST

https://www.youtube.com/playlist?list=PLU8j17gnRm0I



Scan to open the repository

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